Online Appendix

Skill Transferability, Migration, and Development: Evidence from Population Resettlement in Indonesia

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A Data Appendix

For the analysis, we combine information on transmigrant placements with demographic, economic, linguistic, and spatial and agro-climatic characteristics. Table A.1 summarizes the datasets used for this analysis. Each of the datasets is described in detail in the following sections.

A.1 Transmigration Census and Maps

The main source of information on where these transmigrants are placed is the census of transmigration sites established between 1952 and 1998 produced by the Ministry of Manpower and Transmigration (MOMT). The census contains information on the physical locations and names of the transmigration sites, the years they were established, and the number of people and households sent to each site at the time of the initial settlement.[1] From the physical location information, we can identify the villages where these sites reside.

A total of 2,625 sites are listed in the dataset. We manually matched the village names listed there with those listed in the 2000 Indonesian Population Census data and identified 1,702 transmigration villages.[2] This paper analyzes the transmigration sites established during Indonesia's Third and Fourth Development Plan (1979-1988) in the outer islands, which leaves us with 1,021 transmigration villages.

The ministry also produces a set of maps that identify the locations of planned transmigration sites that were ultimately not selected for the program. Outlines of these planned sites – also known as "recommended development areas" (RDAs) were digitally traced using GIS software. We overlay the traced RDA maps onto digital maps of Indonesia's administrative boundaries that were produced by BPS for use in fielding the 2000 and 2010 Household Censuses.[3] We identified a total of 907 "RDA

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[1] Additionally, the database includes information on the type of program implemented in each of these locations and the legal or administrative status of each location.

[2] In most cases, each site corresponds to a village; however, there are some cases where a village comprises multiple sites.

[3] These administrative boundary maps are extremely detailed, with the 2010 map containing over 75,000 polygons identifying the locations of different villages, their names, and the names of the provinces, districts, and subdistricts to which they belong.

villages" in the administrative boundary file that shared any area with the RDA polygons. These villages act as the control villages in our analysis of the average treatment effect (ATE) of the program.[4]

Table A.1: Summary of Datasets

Dataset	Description	Obs. Unit
Transmigrant placement		
Transmigration census	Location of transmigration sites; the number of households and individuals, and years placed in each site.	Transmigration site
Reppprot Maps	1:250,000 Regional Physical Planning Program for Transmigration (Reppprot) maps that include "recommended development area" or RDAs.	
Demographic Variables		
Population Census, 2000	Full dataset: Highest level of schooling, ethnicity, sectoral employment, birth information (year and month, district), district of residence in 1995.	Individual
Intercensal Survey (*SUPAS*), 1985	Intercensal survey with information on migration between districts.	Individual
Population Census, 1980	IPUMS subset (5% of population). Housing characteristics, highest level of schooling, sectoral employment, birth information (year and month, province), province of residence in 1975.	Individual
Economic Variables		
Podes, 2003	Crop types and yield, % of farmland area.	Village
FAO/PriceSTAT	Crop-specific prices.	National
NOAA Light Intensity	Light intensity data, 1992-2010.	30-arc-second grid
Susenas, 2004	Household-level rice productivity, HH-head education level	Household
Linguistic Variable		
World Language Mapping System (WLMS), Ethnologue	Language by ethnic groups and the shared branches of different languages.	
Agroclimatic Variables		
FAO-GAEZ	Potential output (tons/ha) for major crops (see Costinot et al., forthcoming; Nunn and Qian, 2011).	5-arc-minute grid
GIS Map - Dept. Public Works	Village area, distance to coast, roads and others.	
Harmonized World Soil Database	Elevation, ruggedness, soil quality (organic carbon, topsoil characteristics, texture, drainage).	30-arc-second grid
Terrestrial Precipitation and Temperature Data	Rainfall (Matsuura and Wilmott, 2012b) and temperature (Matsuura and Wilmott, 2012a), 1948-1978.	Monthly, interpolated to 0.5×0.5 degree grid

[4]For the analysis of the program ATE, the actual numbers of transmigration and RDA villages included are fewer than 1,021 and 907 respectively because of the island fixed effects and imperfect matching with the rest of the datasets.

A.2 Demographic and Economic Variables

A.2.1 Population Census Data, 2000

Indonesia's 2000 Population Census is a dataset issued by Indonesia's statistical agency, *BPS Statistics* (hereafter, BPS), that was designed as a complete enumeration of the individual members of every household in Indonesia with 100% coverage. However, due to riots and communal violence following the political transition, the population numbers for the provinces of Aceh, Maluku, Papua, and Central Sulawesi had to be estimated (instead of enumerated) by the provincial statistical offices (Surbakti et al., 2000).

The census contains information on the respondents' religion, ethnicity, birth information (year, month, and province), as well as the sector of their employment (if they were working) and their province of residence in 1995. It also includes questions on the respondents' sex, marital status, education, and main activities in the past week. We aggregate the individual-level observations to construct village-level demographic variables and population weights for the similarity indices, and use the individual-level observations to examine occupation choice among the transmigrants.

A.2.2 Village Potential (Podes), 2003

Meanwhile, we construct the agricultural outcome variables using Indonesia's Village Potential (*Podes*) survey, which is also issued by BPS. *Podes* is an administrative census of all villages in Indonesia that collects a rich set of village-level data, and has been conducted approximately every three to four years since 1976. We use the land size and crops information (types and production levels) during the 2001-2002 agricultural season that is available in the 2003 Podes.

A.2.3 FAO/PriceSTAT

We calculate the revenue-weighted average productivity measures using crop prices from the FAO/ PriceStat database. Only crops whose 2001 prices are available are included, namely avocados; bananas; beans, dry; beans, green; cabbages and other brassicas; carrots and turnips; cashew nuts, with shell; cassava; chillies and peppers, green; cinnamon (canella); cloves; cocoa beans; coconuts; coffee, green; cotton lint; cow milk, whole, fresh; cucumbers and gherkins; eggplants (aubergines); garlic; goat meat; groundnuts, with shell; hen eggs, in shell; maize; maize, green; mangoes, mangosteens, guavas; natural rubber; nutmeg, mace and cardamoms; oil palm fruit; onions, dry; oranges; other bird eggs,in shell; palm oil; papayas; pepper (piper spp.); pineapples; potatoes; soybeans; spinach; sweet potatoes; tea; tobacco, unmanufactured; tomatoes; vanilla.

A.2.4 *Susenas* 2004, *SUPAS* 1985, and Population Census Data 1980

We also included three additional national-level datasets published by BPS. First, we use the 2004 *Susenas*, which is a household survey, to estimate the relationship between the education of the household head and rice productivity at the household level. Second, we use the 1985 Intercensal Survey, or *SUPAS* to calculate the inter-district migration flows in the early 1980s. Finally, we use the 5% population subset of the 1980 Indonesia Population Census that are available from IPUMS to construct the pre-1980 district-level characteristics that are included as control variables.

A.2.5 FAO GAEZ data

We downloaded grid-cell level data of potential yield from FAO's Global Agro-Ecological Zones (GAEZ) for wetland rice, dryland rice, cocoa, coffee, palmoil, cassava and maize. To convert grid-level data to village level data, we aggregated across grids using area weights, calculated as the total area of the grid

overlapping with the village, divided by the total village area. We use the same method to convert grid-level data to district-level data at the transmigrants' origin districts.

A.2.6 NOAA Data on Light Intensity, 1992-2010

To proxy for economic activities at the local level, we make use of an innovative technique, developed by Henderson et al. (2012), which uses satellite data on nighttime lights. Daily between 8:30 PM and 10:00 PM local time, satellites from the United States Air Force Defense Meteorological Satellite Program (DMSP) record the light intensity of every 30-arc-second-square of the Earths surface (corresponding to roughly 0.86 square kilometers). DMSP cleans this daily data, dropping anomalous observations, and provides the public with annual averages of light intensity from multiple satellites. After averaging the data across multiple satellites, we obtain annual estimates of light intensity for every 30-arc-second square of the Earths surface from 1992-2010.

Henderson et al. (2012) show that across countries, growth in night-lights (measured as the change in the spatial average digital number of light intensity over time) is linearly related to growth in output. We use the night-lights growth measure in the analysis, but we calculate the changes at the village-level.

A.3 Linguistic Variable: World Language Mapping System (WLMS) and Ethnologue

To construct the linguistic distance measure, we use the *World Language Mapping System* (WLMS) which maps the languages documented in the *Ethnologue* database (Lewis, ed, 2009) to the relevant locations across the world. WLMS maps the languages to locations using the sixteenth edition of the *Ethnologue* database, which contains 6,909 living languages around the world. Its entries for Indonesia contain more than 700 ethnolinguistic groups, including eight ethnolinguistic groups indigenous to Java/Bali.[5]

A critical feature of the *Ethnologue* database is the linguistic trees for each of the 6,909 languages that are available. This linguistic tree, which shows how different languages and dialects are related among the different language families, enables the calculation of the linguistic distance, which is based on the number of shared branches between two languages.

A.4 Spatial, Topographical, and Agroclimatic Variables

Our specifications control for natural endowments by including agricultural and climatic variables. These variables were created from a variety of sources and often were calculated with the assistance of GIS software (ArcView). This section describes those data in detail and how each of the variables were constructed.

A.4.1 Distances and Map Projection

To compute distances correctly, using linear units of measurement (e.g. kilometers), we made use of the Batavia Transverse Mercator (TM) 109 SE projected coordinate system in all of the GIS work. For each village polygon from the administrative boundary shapefile, we constructed the distance to the coast, the nearest river, the nearest road, and major cities using the Euclidean distance tools from ArcView. The shapefiles Indonesia's rivers, roads, major cities, and coast lines were all provided by Indonesia's Department of Public Works (*Departemen Pekerjaan Umum*).

[5]The indigenous Java/Bali ethnicities include, in descending order of population shares in the Outer Islands: Javanese, Sundanese, Balinese, Madurese, Betawi, Tengger, Badui, and Osing.

A.4.2 Slope, Aspect, and Elevation Data

Topographical variables were created using raster data from the *Harmonized World Soil Database* (HWSD), Version 2.0 (Fischer et al., 2008).[6] The raster files are compiled from high-resolution source data and aggregated to 30 arc-second grids.

Elevation data were computed for each village as the average elevation over the entire village polygon, using raster data from HWSD.[7] Slope and aspect data were also recorded for each village and calculated similarly. Variables equal to the average share of each village corresponding to each slope class (0-2%, 2-4%, etc.) were constructed using ArcView.

A.4.3 Ruggedness

A 30 arc-second ruggedness raster was computed for Indonesia according to the methodology described by Sappington et al. (2007), and village-level ruggedness was recorded as the average raster value. The authors propose a Vector Ruggedness Measure (VRM), which captures the distance or dispersion between a vector orthogonal to a topographical plane and the orthogonal vectors in a neighborhood of surrounding elevation planes. To calculate the measure, one first calculates the x, y, and z coordinates of vectors that are orthogonal to each 30-arc second grid of the Earth's surface. These coordinates are computed using a digital elevation model and standard trigonometric techniques.

Given this, a resultant vector is computed by adding a given cell's vector to each of the vectors in the surrounding cells; the neighborhood or window is supplied by the researcher. Finally, the magnitude of this resultant vector is divided by the size of the cell window and subtracted from 1. This results in a dimensionless number that ranges from 0 (least rugged) to 1 (most rugged).[8]

For example: on a (3×3) flat surface, all orthogonal vectors point straight up, and each vector can be represented by $(0, 0, 1)$ in the Cartesian coordinate system. The resultant vector obtained from adding all vectors is equal to $(0, 0, 9)$, and the VRM is equal to $1 - (9/9) = 0$. As the (3×3) surface deviates from a perfect plane, the length of the resultant vector gets smaller, and the VRM increases to 1.

A.4.4 Soil Quality Covariates

We also make use of the HWSD data for soil quality measures. HWSD provides detailed information on different soil types across the world. The HWSD data for Indonesia is taken from information printed in the FAO-UNESCO Soil Map of the World (FAO 1971-1981), a map printed at a 1:5,000,000 scale. For each village, we created the following measures of soil types: percentage of land covered by coarse, medium, and fine soils, percentage of land covered by soils with poor or excessive drainage, average organic carbon percentage, average soil salinity, average soil sodicity, and average topsoil pH.

A.5 Rainfall and Temperature, 1948-1978

The rainfall and temperature data are based on the databased of Matsuura and Wilmott (2012a,b) at the Department of Geography, University of Delaware. They were compiled from a number of sources; for

[6] Data from the HWSD project are publicly available and can be downloaded here: http://www.iiasa.ac.at/Research/LUC/luc07/External-World-soil-database/HTML/index.html?sb=1. The terrain, slope, and aspect database provided by HWSD researchers was compiled from a high-resolution digital elevation map constructed by the Shuttle Radar Topography Mission (SRTM). SRTM data is also publicly available as 3 arc-second digital elevation maps (DEM) (approximately 90 meters resolution at the equator), available here: ftp://e0srp01u.ecs.nasa.gov/srtm/.

[7] The HWSD elevation raster file records the median elevation (in meters) for each 30 arc-second grid of the Earth's surface. The median is computed across space, from the values of all 3 arc-second cells in the SRTM database.

[8] The authors have generously provided a Python script for computing their Vector Ruggedness Measure (VRM) in ArcView. The script and detailed instructions for installation can be found here: http://arcscripts.esri.com/details.asp?dbid=15423.

Southeast Asia, the monthly data come from the Global Historical Climatology Network v2 (GHCN2) database. Matsuura and Wilmott (2012a,b) interpolated these weather data stations to estimate monthly precipitation and temperature to a 0.5×0.5 degree (or 55 km) resolution grid. Then, for the districts in the dataset, we averaged these numbers for the period of 1948-1978 to obtain the pre-determined measures of rainfall and temperature.

A.6 Constructing Key Variables

Table A.2 summarizes the data sources for our key outcome variables and regressors. The specific formulas to calculate each of these variables are provided in the main text.

Table A.2: Key Variables

Variables	Description	Data source*
Demographic variables		
Log population density	Log of population/total village area (m^2).	2000-PopCen (pop.), GIS (total area)
% born in Java/Bali	% villagers born in Java/Bali	2000-PopCen
Transmigrant ethnicity share	% villagers of Java-based/Balinese ethnicities	2000-PopCen
Average years of schooling	Calculated based on the highest educational attainment.	2000-PopCen
Economic outcomes		
Sectoral employment choice	Individual choice of sectoral employment	2000-PopCen
Log rice productivity	Log of rice tonnage per hectare in the village	2003-Podes (rice yield, farm area)
Revenue-weighted average yield	Average village-level yield of all crops produced weighted by the share of each crop's revenue in the village	2003-Podes (crop yield, farm area), FAO (crop prices)
Log light intensity growth	The log growth of light intensity in the village, 1992-2010, conditional on any light coverage in 1992 and 2010	NOAA
Similarity indices		
Agrosimilarity indices	Agroclimatic similarity between two locations, aggregated to the village-level using population weights	2000-PopCen (pop. weights), Section A.4 for agroclimatic chars
Linguistic similarity	Linguistic similarity between two ethnoliguistic groups aggregated to the village-level using population weights	2000-PopCen (pop. weights), *Ethnologue* (linguistic distance)

Note: *FAO = FAO/PriceSTAT; GIS = GIS Map; NOAA = National Oceanic and Atmospheric Administration's (NOAA) night lights data; PopCen = Population Census; Podes = Village Potential.

B Additional Figures and Tables

This section provides additional tables and figures referenced in the paper.

B.1 Developing the ATE Identification Strategy

This subsection describes the reweighting procedure used in estimating average treatment effects (ATE) in Table 12 of the paper. First, we predict the probability of being a Transmigration village

$$\mathbb{P}(T_j = 1) = \Lambda(\mathbf{x}_j' \widehat{\boldsymbol{\zeta}}) \tag{B.1}$$

where $\Lambda(\cdot)$ is a logit function. The estimates $\widehat{\boldsymbol{\zeta}}$ reported in Table B.1 are indicative of sequential site selection among eligible settlement areas. In particular, treated villages (i) are smaller in area, (ii) are at lower altitude, (iii) have more acidic soils, (iv) have better drainage, and (v) are closer to major roads. The covariates explain about one-third of the variation in site selection, and the estimated propensity scores, $\widehat{P}_j$, for treated and control villages exhibit considerable overlap (see Figure B.4).[1] Using the estimates of $\boldsymbol{\zeta}$, we reweight control village j according to its estimated odds of having been a Transmigration site, $\widehat{\kappa} = \widehat{P}_j/(1 - \widehat{P}_j)$, where $\widehat{P}_j$ is the predicted probability from equation (B.1). This reweighting of control villages effectively rebalances the sample as if planners in 1979 randomly chose half of the initial potential settlements. Without the $\widehat{\kappa}$ weights, more than half of the site selection variables exhibit large and statistically significant differences across treated and control villages. With the weights, that share falls to less than 10 percent. By removing observable site selection differentials, we get closer to a causal interpretation of θ in equation (9) in the paper.

B.2 Additional Robustness Checks

Table B.4 reports the results of the robustness checks for our main rice productivity result discussed in Section 6.1 in the paper. The results are unchanged from the baseline in row 1 when controlling for the log number of total transmigrants placed in the initial year of settlement (row 2) or indicators for the year of settlement (row 3). Row 4 controls for 124 fixed effects for province $\times$ year of settlement. This very demanding specification cuts the estimate in half, but we still find a statistically and economically meaningful positive effect. This is reassuring given that these FE account for, among other things, time-varying differences in the quality of local institutional support for the transmigrants in those important first months after arrival. We partially address aggregation bias in row 5 by controlling for the share of the population from Java/Bali as well as overall log population density. We consider a second, more direct check for aggregation bias below. We also find no substantial change when controlling for a third-order polynomial in the latitude and longitude of the village in row 6. This offers a demanding and flexible way to account for spatial heterogeneity in unobservable determinants of productivity.

Rows 7-10 show robustness to different ways of constructing the similarity index. Rows 7 and 8 use different distance metrics and choice of functional forms. Row 9 uses population weights restricted to migrants arriving before 1995, and row 15 restricts to Java/Bali migrants who were at least 30 years old in the year 2000 (and hence eligible to be relocated through the program). These changes address concerns that our baseline π_{ij} terms that include all Java/Bali migrants (calculated using birth locations in the 2000 Census) may not be capturing the original transmigrants. In results available upon request, we find that the key results for other development outcomes (Table 6 in the paper) are similarly robust.

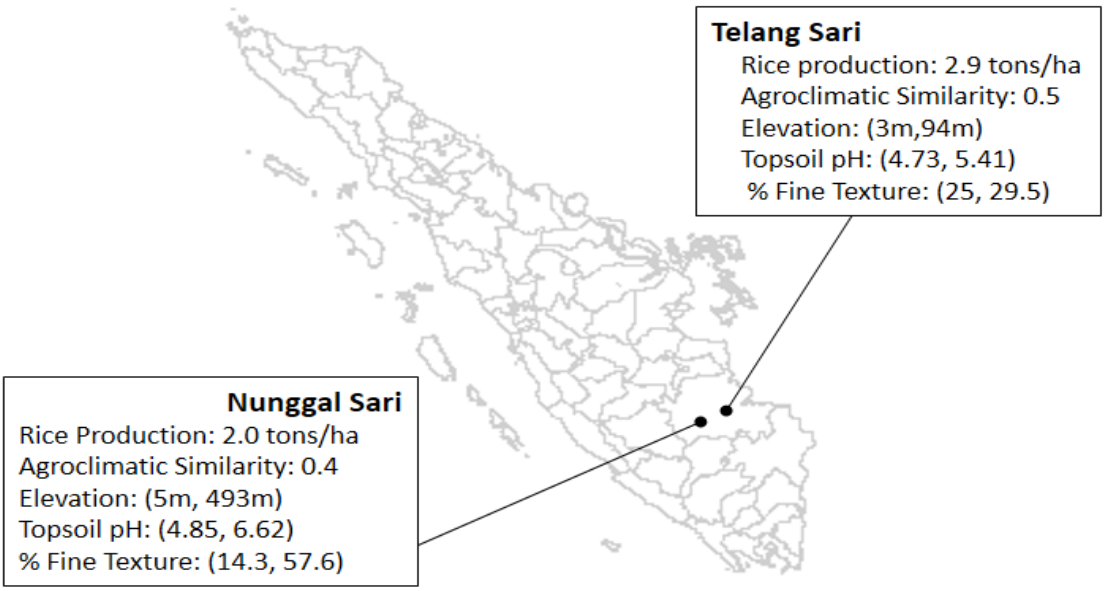
In Table B.3, we address additional concerns about aggregation bias using auxiliary microdata. By regressing village-level outcomes on key similarity regressors that only apply to a subset of villagers

[1] Column 2 extends the sample to include villages that are within 10 kilometers of treated and almost treated villages yields similar estimates. However, we explain less of the variation in site selection, and it becomes more difficult to distinguish treated from control villages when including those areas adjacent to but not within the original boundaries of the eligible settlements. This suggests that our model captures much of the same local, small-area variation identified by planners.

(transmigrants), we risk misinterpreting the relationship between similarity and productivity.[2] We use the best available household-level survey data from a small sample of 74 Transmigration villages. Unfortunately, this dataset does not include migration data, but it does report ethnicity of the household head. We therefore construct an agroclimatic similarity index using ethnic weights instead of birth locations. Estimating a household-level analogue to our main village-level estimating equation, we find that agroclimatic similarity has a positive effect on farm-level rice yields that is qualitatively and quantitatively very similar to our main estimate of γ_a. This is supportive evidence that the main productivity effects we identify in the paper are driven by transmigrants rather than natives.

Figures

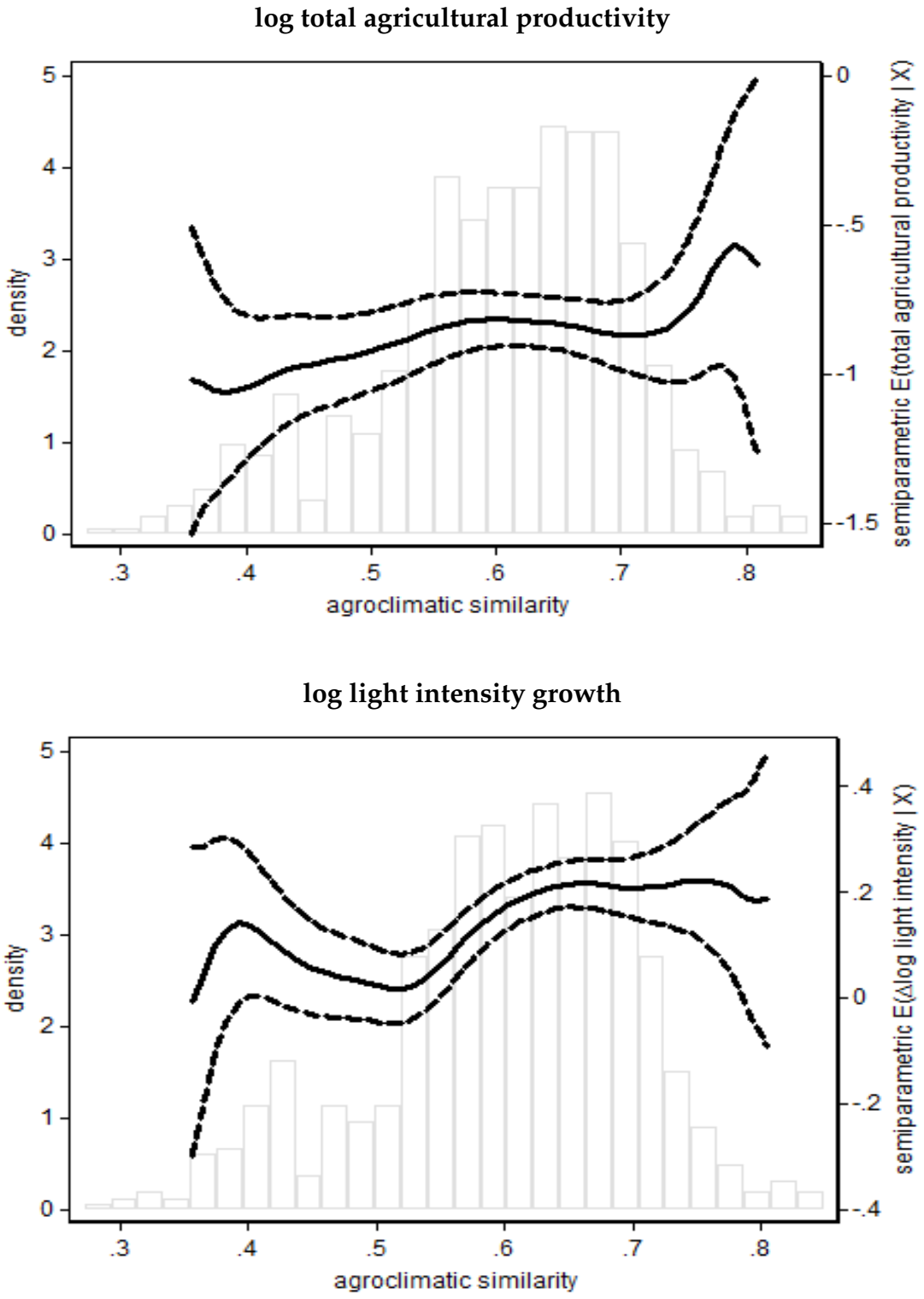
Figure B.1: A Case Study to Explain Our Empirical Strategy



Notes: Values in parentheses correspond to (destination village, origin district), where "origin district" is the district that sent the largest share of transmigrants to the destination village. Both transmigration villages are in Musi Banyuasin District, South Sumatera. For Telang Sari, 23% of its Java/Bali migrants were born in Kebumen, in Central Java. For Nunggal Sari, 38% of its Java/Bali migrants were born in Karanganyar, in Central Java. The standard deviations are: agroclimatic similarity (0.17), Elevation (268m), pH (0.73), Fine Texture (16.1%).

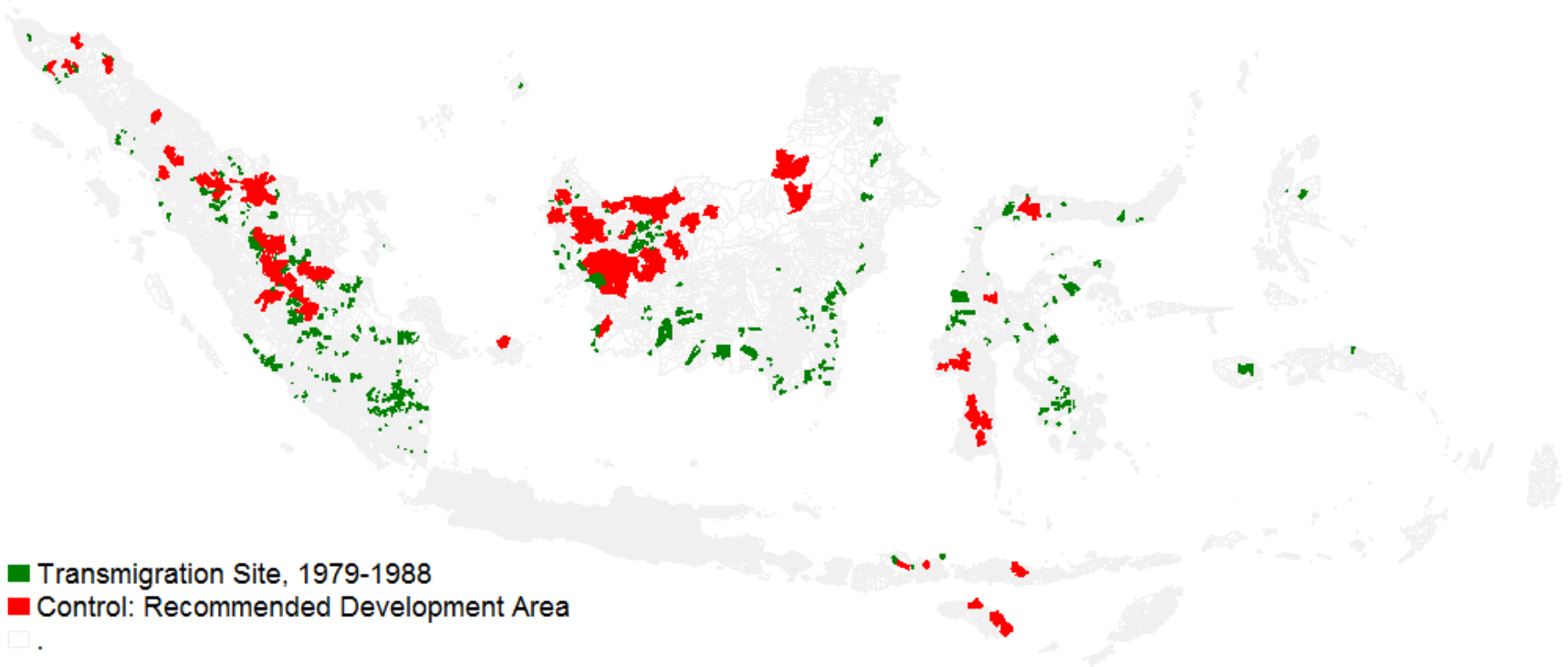
[2]It is also important to note that the estimates of γ_a are largely unchanged when multiplying our village-level agricultural productivity outcomes by the share of the population working in the given sector as reported in the 2000 Population Census (i.e., multiplying total agricultural productivity by the share working as farmers, rice and food crop productivity by the share working as food crop farmers, and cash crop productivity by the share working in cash crops).

Figure B.2: Semiparametric Regressions for Other Development Outcomes



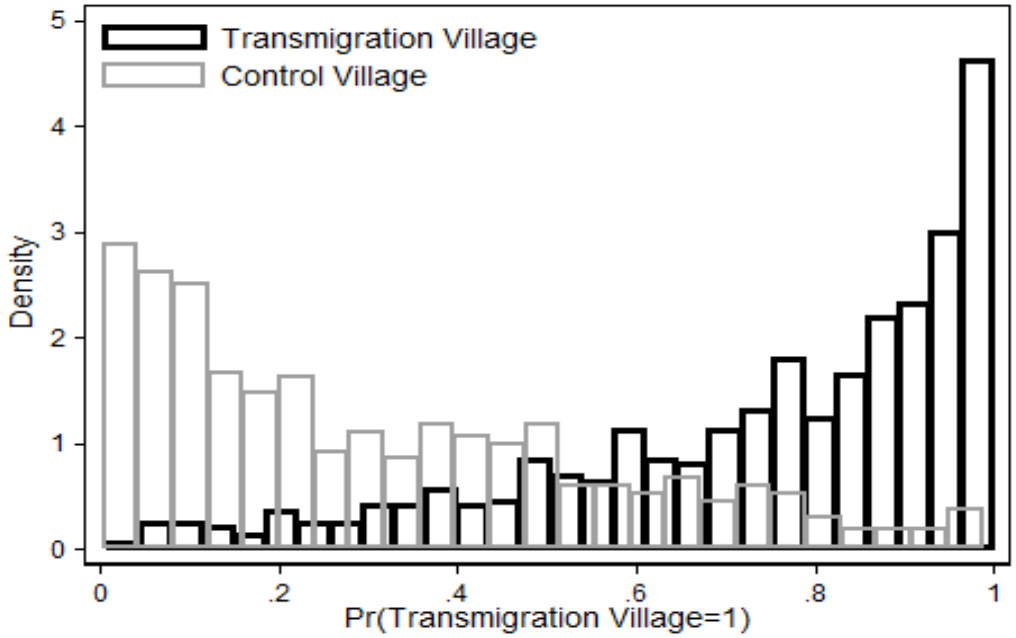
Notes: This is based on semiparametric Robinson (1988) extensions of the main parametric specification in column 3 of Table 5. The dashed lines correspond to 90% confidence intervals based on clustering of standard errors at the district level. The local linear regressions use an Epanechnikov kernel and a bandwidth of 0.05. The histogram captures the distribution of standardized agroclimatic similarity. The top 5 and bottom 5 villages are trimmed in the figure for presentational purposes.

Figure B.3: Map of Transmigration and Control Villages



Notes: Each colored location on the map corresponds to a Transmigration village or a control/RDA village. The white areas outlined in grey are neither Transmigration nor control villages.

Figure B.4: Estimated Propensity Scores



Notes: This figure plots the distribution of estimated probabilities of site selection based on the estimates in column 2 of Table B.7.

Tables

Table B.1: Top 5 Crops by Predicted Revenue in 2002: Transmigration Villages

	Ranking				
Crop	(1)	(2)	(3)	(4)	(5)
rice	0.292	0.217	0.122	0.064	0.036
palm oil	0.261	0.062	0.029	0.019	0.008
rubber	0.131	0.162	0.070	0.023	0.018
cassava	0.069	0.073	0.071	0.043	0.047
cocoa	0.058	0.053	0.019	0.017	0.008
coffee	0.030	0.022	0.021	0.032	0.031
pepper	0.025	0.006	0.014	0.005	0.006
maize	0.021	0.069	0.062	0.062	0.073
banana	0.017	0.049	0.064	0.047	0.021
soybean	0.016	0.014	0.032	0.031	0.025
groundnut	0.016	0.031	0.066	0.055	0.045
coconut	0.012	0.022	0.030	0.053	0.038
chili pepper	0.010	0.017	0.039	0.045	0.035
orange	0.009	0.027	0.027	0.019	0.018
mango, guava, mangosteen	0.009	0.017	0.026	0.027	0.025
cloves	0.006	0.006	0.005	0.012	0.004
cashew	0.004	0.005	0.004	0.003	0.006
sweet potato	0.004	0.005	0.013	0.017	0.016
cinnamon	0.003	0.003	0.004	0.004	
greenbean	0.001	0.012	0.022	0.044	0.042
spinach	0.001	0.006	0.008	0.006	0.010
nutmeg, cardamom	0.001				0.001
garlic	0.001				
avocado	0.001	0.001		0.001	0.001
cucumber	0.001	0.010	0.021	0.016	0.017
tomato			0.001	0.009	0.001
cotton		0.001			
onion		0.004		0.001	0.004
pineapple				0.001	0.003
papaya		0.001		0.006	0.003
eggplant			0.008	0.013	0.018

Notes: This table shows the percentage of Transmigration villages with the given crop as their first, second, third, fourth, or fifth most valuable product as measured by the total output in the 2001 growing season valued at the national price given in FAO PriceSTAT series. The figures are only defined over villages reporting nonmissing and nonzero agricultural output in any crop for which we have price data.

Table B.2: Agroclimatic Similarity and Village-Level Demographics

Dependent Variable	agroclimatic similarity
	(1)
Male (%), Java/Bali Migrants	-0.004
	(0.052)
Avg. Age, Java/Bali Migrants	0.002
	(0.002)
Post-Primary School (%), Java/Bali Migrants	0.006
	(0.029)
Primary School (%), Java/Bali Migrants	0.043
	(0.042)
Male (%), Other	-0.070
	(0.247)
Avg. Age, Other	-0.002
	(0.002)
Post-Primary School (%), Other	-0.023
	(0.072)
Primary School (%), Other	0.021
	(0.099)
Number of Villages	814
Predetermined Natural Advantages **X**	Yes
Island Fixed Effects	Yes
R^2	0.479

Notes: */**/*** denotes significance at the 10/5/1 percent level. This table regresses agroclimatic similarity on the demographic characteristics of Transmigration villages reported in the 2000 Population Census and broken down by Java/Bali-born and non-Java/Bali-born population. Each regression includes the predetermined natural advantages as well as island fixed effects. Standard errors in parentheses allow for unrestricted spatial correlation between all villages within 150 kilometers of each other (Conley, 1999).

Table B.3: Individual-Level Rice Productivity Regression (74 Transmigration Villages)

Ethnicity	**All**	**Java/Bali**	**Outer Islands**
	(1)	**(2)**	**(3)**
agroclimatic similarity	0.062	0.150**	0.123
	(0.053)	(0.073)	(0.199)
Number of Households	546	449	97

Notes: */**/*** denotes significance at the 10/5/1 percent level. Individual-level regressions of log rice output per hectare for individuals (household heads) living in a random sample of 74 Transmigration villages in a nationally representative household survey (*Susenas*) conducted in 2004. Although we do not observe migration histories (i.e., district of birth), we can use the observable individual (household head) ethnicity to construct a proxy for the individual's agroclimatic similarity based on the prevailing origins of the given ethnic population in the village. Agroclimatic similarity is defined at the individual-level based on an origin-weighted average of the ethnicity-specific agroclimatic similarity prevailing across individuals in the village as observed using the full 2000 Population Census. All regressions include predetermined village-level controls. Standard errors are clustered by district.

Table B.4: Robustness: Rice Productivity

	agroclimatic similarity
1. Baseline Specification	0.204 (0.064)***
2. Total Transmigrants Placed in Initial Year	0.205 (0.064)***
3. Year of Settlement Fixed Effects	0.200 (0.063)***
4. Province $\times$ Year of Settlement Fixed Effects	0.114 (0.065)*
5. Controlling for Java/Bali-born Pop. Share and Overall Pop. Density	0.211 (0.063)***
6. 3rd Degree Polynomial in Latitude/Longitude	0.193 (0.077)**
7. Alternative Normalization of Agroclimatic Similarity Index	0.192 (0.060)***
8. Euclidean Distance in Agroclimatic Similarity Index	0.161 (0.086)*
9. Only pre-1995 Java/Bali Immigrants in Agroclimatic Similarity Index	0.206 (0.067)***
10. Only Java/Bali-born age >30 in Agroclimatic Similarity Index	0.212 (0.060)***

Notes: */**/*** denotes significance at the 10/5/1 percent level. Each row corresponds to a separate regression of log rice productivity on agroclimatic similarity, predetermined village-level control variables, and island fixed effects unless noted otherwise. Agroclimatic similarity is normalized to have mean zero and a standard deviation of one. Row 1 is our baseline specification from column 1 of Table 5 in the paper. In each subsequent row, we make the single change in specification noted in the row description. Otherwise, all other aspects of the estimating equation are as in the baseline. Standard errors in parentheses allow for unrestricted spatial correlation between all villages within 150 kilometers of each other (Conley, 1999).

Table B.5: Which Components of Agroclimatic Similarity Matter (Most)?

	(1)	(2)	(3)	(4)	(5)
agroclimatic similarity	0.204				
	(0.064)***				
topographic similarity		0.203			0.327
		(0.088)**			(0.160)**
soil similarity			0.158		-0.142
			(0.079)**		(0.156)
climatic similarity				-0.085	-0.110
				(0.056)	(0.058)*
Number of Villages	600	600	600	600	600
R^2	0.252	0.252	0.247	0.243	0.258

Notes: */**/*** denotes significance at the 10/5/1 percent level. Each column corresponds to a regression of log rice productivity on agroclimatic similarity, predetermined village-level control variables, and island fixed effects. Log rice productivity is tons of output per hectare in the 2001-2 growing season. Revenue weighted log agricultural productivity is an overall measure of agricultural output in tons per hectare that each crop by the (potential) revenue of total output valued at national prices. Both similarity measures are normalized to have mean zero and a standard deviation of one. Standard errors in parentheses allow for unrestricted spatial correlation between all villages within 150 kilometers of each other (Conley, 1999).

Table B.6: Agroclimatic Similarity and Crop Choice

	agroclimatic similarity	Dep. Var. Mean
Dependent Variable	Panel A: **Food Crops**	
any food crop production	0.089	0.888
	(0.022)***	
any rice production	0.088	0.737
	(0.019)***	
any cassava production	0.037	0.442
	(0.037)	
any maize production	0.037	0.441
	(0.038)	
	Panel B: **Cash Crops**	
any cash crop production	-0.030	0.909
	(0.020)	
any rubber production	0.010	0.410
	(0.031)	
any palm oil production	0.018	0.382
	(0.022)	
any coffee production	0.055	0.184
	(0.017)***	
any cocoa production	0.020	0.154
	(0.009)**	

Notes: */**/*** denotes significance at the 10/5/1 percent level. Each row corresponds to a separate regression for the given dependent variable using the same baseline specification applied in column 1 Table 5 in the paper. All dependent variables are as observed in the 2001-2 growing season. For each crop we report the extensive margin relationship between agroclimatic similarity and the (linear) probability that the crop is grown in the village. *any food* and *any cash* equal one if the village grows any of a large list of crops of the given type (see the notes to Table 6 in the paper). Agroclimatic similarity is normalized to have mean zero and a standard deviation of one. All regressions include predetermined village-level control variables and island fixed effects. Standard errors in parentheses allow for unrestricted spatial correlation between all villages within 150 kilometers of each other (Conley, 1999).

Table B.7: Determinants of Site Selection

Treated/Control Radius	0 km	10 km
	(1)	**(2)**
log village area, Ha	-0.103	-0.028
	(0.019)***	(0.014)**
% w/ slope between 0-2%	0.006	0.002
	(0.002)***	(0.001)**
Vector Ruggedness Measure	-0.164	-0.027
	(0.115)	(0.076)
log altitude, m^2	-0.026	-0.018
	(0.009)***	(0.008)**
Organic Carbon (%)	-0.020	-0.010
	(0.006)***	(0.007)
Topsoil Sodicity (ESP) %	0.086	0.006
	(0.093)	(0.065)
Topsoil pH (-log(H+))	-0.141	-0.155
	(0.051)***	(0.041)***
Coarse texture soils (%)	-0.033	0.108
	(0.226)	(0.214)
Very poor or poor drainage (%)	0.073	-0.032
	(0.085)	(0.081)
Imperfect drainage soils (%)	-0.231	-0.132
	(0.138)*	(0.100)
Avg. rainfall, 1948-1978	-0.001	-0.001
	(0.001)	(0.001)**
Avg. temp (Celcius), 1948-1978	-0.022	0.002
	(0.014)	(0.012)
Distance to Nearest Major Road	-0.300	-0.255
	(0.157)*	(0.165)
Distance to Nearest Coast	-0.060	-0.065
	(0.038)	(0.029)**
Distance to Nearest River	-0.011	-0.023
	(0.022)	(0.013)*
Distance to District Capital	0.025	0.014
	(0.028)	(0.017)
N	1470	5032
Pseudo R^2	0.366	0.284
Log Likelihood	-641.9	-2109.1
LR χ^2	365.1	143.9

Notes: */**/*** denotes significant at the 10/5/1 percent significance levels. This table reports average marginal effects. The dependent variable is a binary indicator equal to one if the village is located within 0 or 10 kilometers of a Transmigration site. Standard errors clustered by district in parentheses.

C Reassigning Transmigrants to Maximize Aggregate Production

The problem of finding the optimal allocation of transmigrants to destination villages is an example of the generalized assignment problem (GAP), a problem in computer science and operations research. This problem has been shown to be NP-Hard (Fischer et al., 1986).[1] Intuitively, solutions to the problem require a search over the space of partitions of the set of transmigrant individuals, which we can represent by $\{1, 2, ..., N\}$. A partition would divide this set into subsets consisting of individuals assigned to each destination location. Without constraints, a brute force algorithm for assigning people to places would have to check all possible different partitions for optimality. It has been shown that the number of partitions of a collection of N objects is given by B_N, the Bell number for a set of size N. The number of partitions of a set of increases faster than e^N; the first few Bell numbers are $B_1 = 1$, $B_2 = 2$, $B_3 = 5$, $B_4 = 15$, $B_5 = 52$, $B_6 = 203$, $B_7 = 877$, $B_8 = 4140$, $B_9 = 21147$, and $B_{10} = 115975$.

However, we can obtain a fast approximation to the optimal solution by using a "greedy" assignment algorithm. The algorithm first orders villages by the total number of spaces for Transmigrants, from smallest to largest. This gives us an ordering of $v \in \{1, 2, ..., N\}$.[2] Next, we maximize similarity in village 1 by choosing people from origin locations with the smallest agricultural distance to village 1. Using the remaining unassigned people, we maximize similarity in village 2, and we continue. Our program to do this is computationally efficient and provides us with a new set of agroclimatic similarity indices, $\widetilde{A}_1$, $\widetilde{A}_2$, ..., $\widetilde{A}_N$.

Maximizing similarity across villages is of first order importance to rice production, because for this exercise, we assume that log rice output per hectare in village v is given by:

$$\log Y_v = x_v'\beta + \theta A_v$$

where β' and θ are known parameters, obtained from Table 5, Column 1.[3] Fixing x_v, we can predict the new rice output per hectare in village v as follows:

$$\log \widetilde{Y}_v - \log Y_v = \theta \left(\widetilde{A}_v - A_v \right)$$

where $\widetilde{A}_v$ is village v's optimized similarity index. We construct the new output per hectare by rearranging:

$$\widetilde{Y}_v = \exp \left\{ \widehat{\theta} \left(\widetilde{A}_v - A_v \right) \right\} Y_v$$

Multiplying this result by H_v, village v's total hectares of cultivation, gives us the new tons of rice per hectare that would have been produced.

We can compare the total rice production across all villages to our counterfactual rice production if villages had been assigned transmigrants in a way that maximized similarity. This gives us:

$$\Delta = \sum_{v=1}^{V} \widetilde{Y}_v H_v - \sum_{v=1}^{V} Y_v H_v$$

From this calculation, we found that total aggregate rice production could have been 27.5% higher if individuals had been assigned in a more optimal manner.

[1] GAP has many applications, including vehicle routing, assignment of software development tasks to programmers, assigning jobs to computers in computer networks, and designing communication networks

[2] Note that our greedy assignment algorithm starts by maximizing similarity in the smallest village. We tried randomizing the order of villages in which similarity was maximized, and found that we could do no better than this ordering choice. However, total aggregate output always improved; in 100 different randomly assigned orderings, the range of output improvements was 16.3% to 27.4%.

[3] Note that we assume that the only dimension of similarity that is affected by the reassignment is agroclimatic similarity.

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